

The logic of arguments - Set 2 - Solutions

Specification Arg1-Arg4.

Answer key

Question	Answer	Topic
1	D	The logic of arguments
2	B	The logic of arguments
3	B	The logic of arguments
4	E	The logic of arguments
5	A	The logic of arguments
6	B	The logic of arguments
7	A	The logic of arguments
8	A	The logic of arguments
9	G	The logic of arguments
10	B	The logic of arguments
11	E	The logic of arguments
12	H	The logic of arguments

Worked solutions

Question 1 LGC-21 · The logic of arguments · spec Arg3

Consider the statement S : **for all real x , $x^2 - 6x + 10 > 0$.** Which of the following is true?

Answer: D

Completing the square shows the expression is at least 1 for every x .

Question 2 LGC-22 · The logic of arguments · spec Arg2

For real numbers a, b : P : ' $ab = 0$ '; Q : ' $a = 0$ '. Which one of the following is true?

Answer: B

$a = 0 \Rightarrow ab = 0$, so P is necessary for Q . But $ab = 0$ could be because $b = 0$ with $a \neq 0$, so P is not sufficient.

Question 3 LGC-23 · The logic of arguments · spec Arg1

Consider the statement: **if a and b are irrational, then $a + b$ is irrational.** Which of the following pairs is a counterexample?

Answer: B

A counterexample needs the hypothesis true (both irrational) and the conclusion false ($a + b$ rational):
 $\sqrt{2} + (-\sqrt{2}) = 0$.

Question 4 LGC-24 · The logic of arguments · spec Arg4

Let S be the statement: **every even number greater than 2 can be written as the sum of two primes.** A single counterexample to S would be

Answer: E

To refute 'for all n , $P(n)$ ' you exhibit one n with $P(n)$ false.

Question 5 X2-24 · The logic of arguments · spec Arg2

For a real number x , P : ' $x > 0$ '. Q : ' $x^3 > 0$ '. Which one of the following is true?

Answer: A

Cubing preserves sign, so $x > 0 \Leftrightarrow x^3 > 0$.

Question 6 LGC-05 · The logic of arguments · spec Arg3

Which of the following statements is/are true?

- I For every real number x , there exists a real number y such that $y > x$.
- II There exists a real number y such that for every real number x , $y > x$.
- III For every real number x , there exists a real number y such that $xy = 1$.

Answer: B

I is true (take $y = x + 1$). II claims a single largest real number exists: false. III fails for $x = 0$.

Question 7 LGC-06 · The logic of arguments · spec Arg1

A teacher says: '**You will pass only if you revise.**' Which of the following statements means the same thing?

Answer: A

'A only if B' means ' $A \Rightarrow B$ ': passing implies revising. It does not promise that revising guarantees a pass.

Question 8 LGC-12 · The logic of arguments · spec Arg3

Consider the statement: ****there exists a real number k such that for all real x , $x^2 + kx + 1 > 0$.**** Which of the following is true?

Answer: A

Need discriminant $k^2 - 4 < 0$, i.e. $-2 < k < 2$. So some (not all) values of k work; $k = 0$ gives $x^2 + 1 > 0$. ($k = 2$ gives $(x + 1)^2$, which is 0 at $x = -1$.)

Question 9 LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I** For all x and all y , if $x < y$ then $x^2 < y^2$.
- II** For all x there exists y such that $x < y$ and $x^2 < y^2$.
- III** There exist x and y with $x < y$ and $x^2 > y^2$.

Answer: G

I fails for $x = -2$, $y = 1$. II: take $y = |x| + 1$. III: $x = -2$, $y = 1$ works.

Question 10 LGC-18 · The logic of arguments · spec Arg4

A statement reads: ****for every positive integer n , there is a prime p with $n < p < 2n$.**** Which of the following is its negation?

Answer: B

Negate the quantifiers in turn: 'for every n , there exists p with ...' becomes 'there exists n such that for every prime p , not ...', i.e. no prime lies strictly between n and $2n$.

Question 11 LGC-20 · The logic of arguments · spec Arg1

Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I** It is not raining, or the ground is wet.
- II** If the ground is not wet, then it is not raining.
- III** If the ground is wet, then it is raining.

Answer: E

' $A \Rightarrow B$ ' is equivalent to 'not A , or B ' (I) and to its contrapositive (II). III is the converse, which is not equivalent.

Question 12 LGC-09 · The logic of arguments · spec Arg1

Consider the statement S : **for all real x , if $x > 0$ then $x + \frac{1}{x} \geq 2$.** Which of the following statements is/are true?

- I** S is true.
- II** The converse of S (for all real x , if $x + \frac{1}{x} \geq 2$ then $x > 0$) is true.
- III** The contrapositive of S is true.

Answer: H

For $x > 0$, $x + \frac{1}{x} - 2 = \frac{(x-1)^2}{x} \geq 0$, so S is true, and so is its contrapositive (III). For the converse: if $x < 0$ then $x + \frac{1}{x} < 0 < 2$, so $x + \frac{1}{x} \geq 2$ forces $x > 0$ (II true).